

Introduction to IoT

Definitions, Characteristics, Applications

What is IoT?

- ✓ Internet of Things is the **network of smart physical objects**
 - physical objects (e.g. devices, vehicles, buildings, etc.) **embedded with** sensors, computation unit, memory unit, power source, and network connectivity,
 - which enables the physical object to **collect and exchange data**,
 - **analyze** the collected data to extract new insight and **respond** accordingly.
- ✓ Goal of IoT is to “**connect the unconnected**”
 - “Things” or “objects” **that were not supposed to be connected** to the Internet



➤ IoT did the **technology transition** in traditional computer networks

Cont...

- IoT is the Unifications of technologies:
 - ✓ Embedded Systems
 - ✓ Low Power and Low Rate Network
 - ✓ Internet
 - ✓ Cloud Computing
 - ✓ Data Analytics
 - ✓ Bigdata
 - ✓ Edge Intelligence
 - ✓ Network Security and Data Security
 - ✓ Software Defined Networks
 - ✓ Etc.
- Alternate Definition:

“The Internet of Things (IoT) is the **network of physical objects** that contain embedded technology to **communicate** and **sense** or **interact** with their **internal states** or the **external environment**.” – Gartner Research*

* <https://www.gartner.com/en/information-technology/glossary/internet-of-things>

Brief History of Initial Phase of IoT

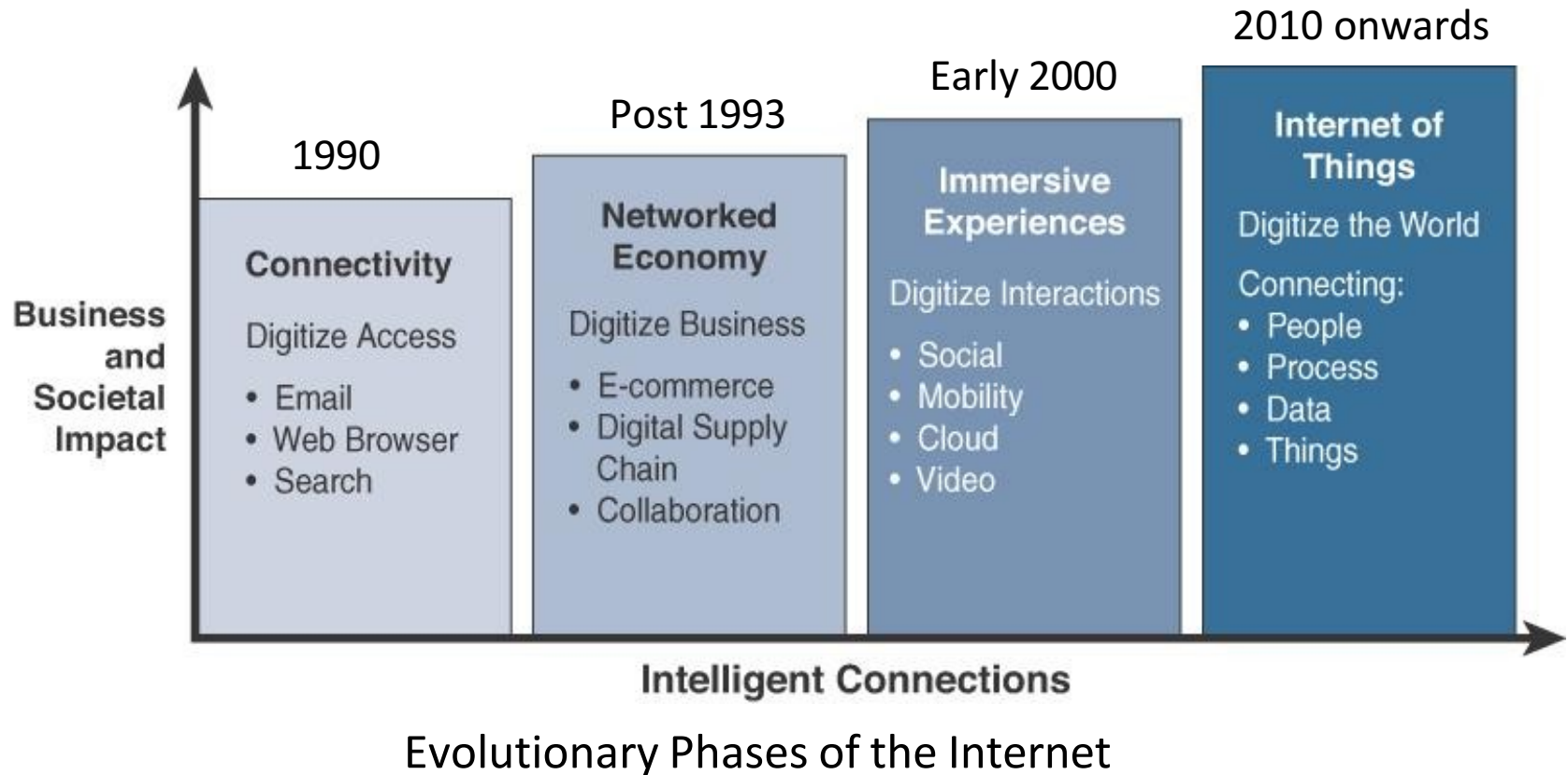
- The term "**Internet of Things**" was likely coined by **Kevin Ashton** of Procter & Gamble, later MIT's Auto-ID Center, **in 1999**.
 - Kevin Ashton said –
 - “In 20th century, **computers were brains without senses** —they only knew what we told them.”
 - “Now in 21st century, **computers are sensing things for themselves!**”
- **Early 1980s** at the Carnegie Melon University, a group of students created a way to get their campus **Coca-Cola vending machine** to report on its contents **through a network** in order to save them the trek if the machine was out of Coke.

Brief History of Initial Phase of IoT

- **In 1990**, John Romkey, developer of the first TCP/IP stack for IBM PC in 1983, connected a toaster to the internet for the first time.
- **In 1991**, a group of students at the University of Cambridge used a web camera to report on coffee available in their computer labs coffee pot.
- At the **beginning of the 21st Century**, LG Electronics introduced the world's first refrigerator connected to the internet

Brief History of Initial Phase of IoT

- The popularity of IoT did not accelerate until 2010/2011 and reached mass market from 2013-14.
- Definition of the IoT has also evolved over time.



Benefits of IoT

- Real-Time Monitoring

- IoT sensors can monitor equipment and processes in real-time, providing valuable sensed data that can be used to streamline operations, reduce waste, and increase output.

- Automation of Processes

- Machines can assemble parts with more precision and speed, resulting in fewer errors during assembly
- Robots can very rapidly detect faults that may not be detected by the human eye

- Improved Efficiency or Productivity

- IoT helps to enhance productivity by streamlining processes, automating tasks, and providing real-time data insights.

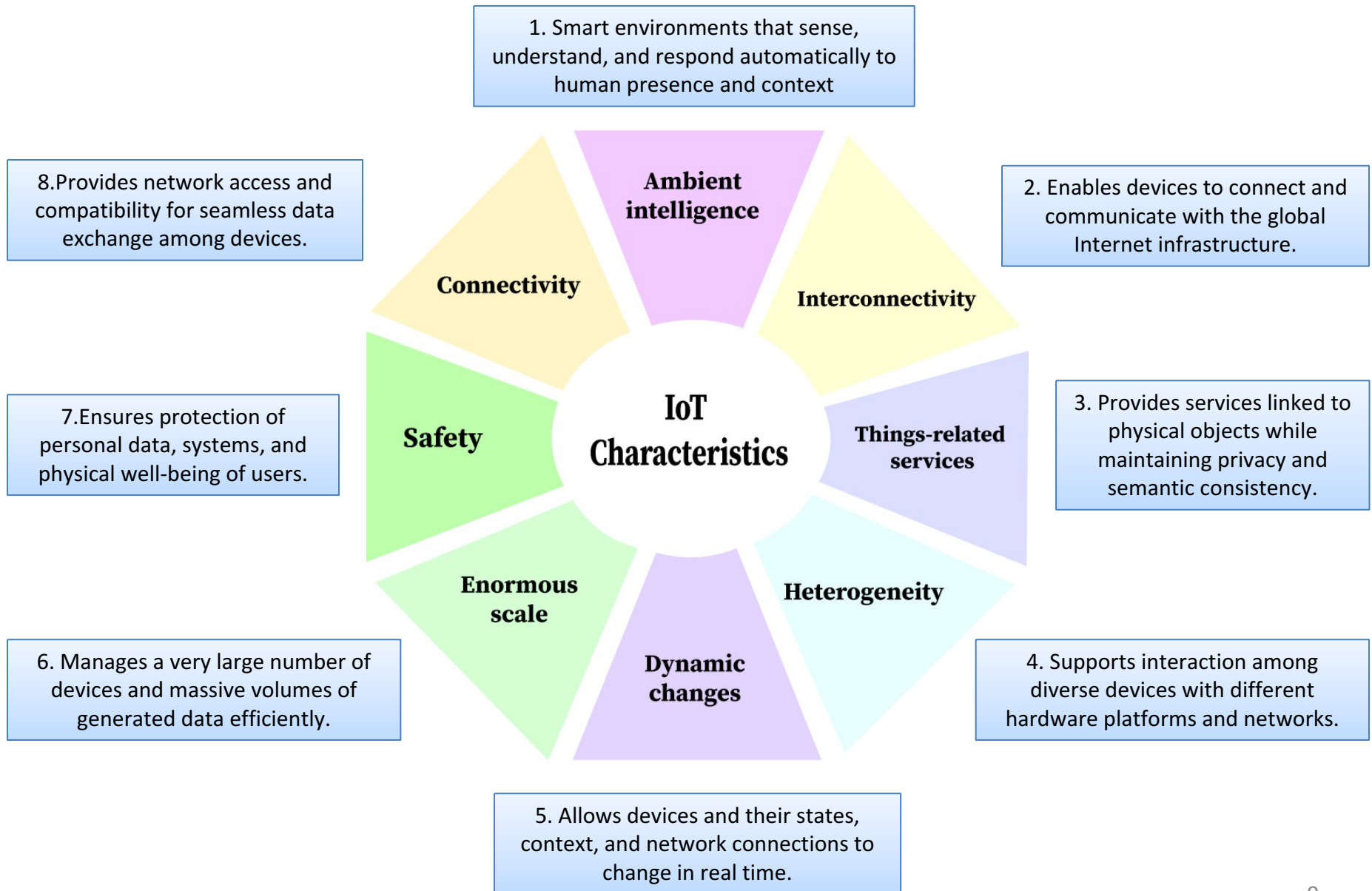
- Predictive Maintenance

- Continuous monitoring of systems and processes to identify key indicators of problems before they result in downtime or system failure

Benefits of IoT

- Improved or New Insights
 - IoT generates vast amounts of data that can be analyzed to identify trends, inefficiencies, and areas for improvement. Organizations can leverage these insights to enhance productivity and operational effectiveness.
- Cost Reduction
 - When an organization can improve system uptime, automate processes, reduce the risk of failure and gain insights that support better decision making, and reduce resource usage, the result is efficiency and cost savings
- Optimized Work Environments
 - IoT technology can help create more comfortable and efficient workspaces
- Adaptability
 - The ability to adapt to new business requirements, customer needs, and changing conditions, or scale the deployment in response to business growth or customer requirements

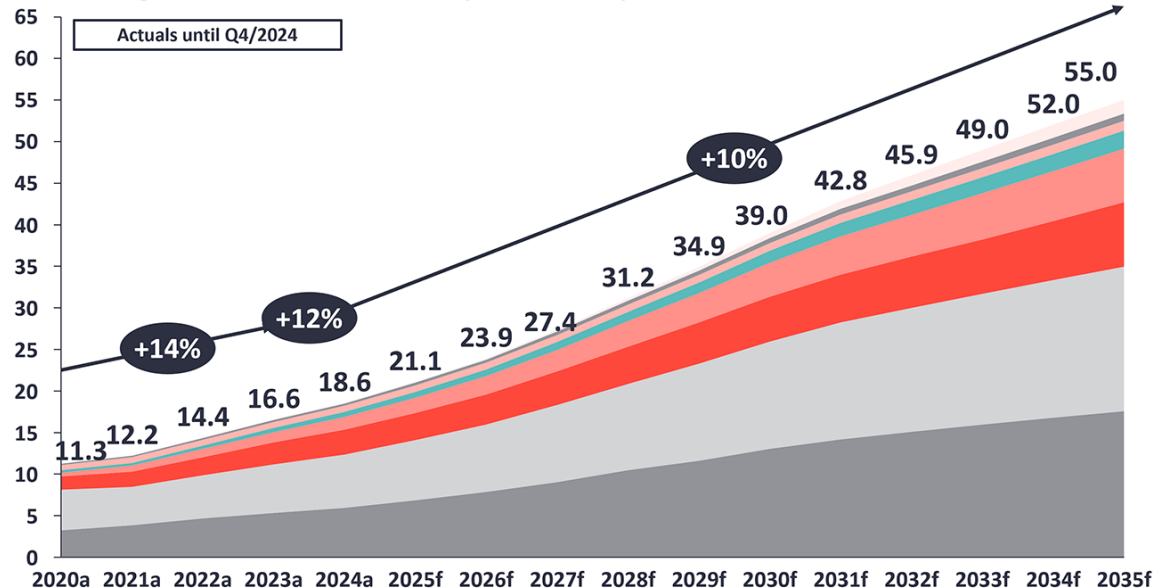
Characteristics of IoT



Growth of IoT Devices

Global IoT market forecast (in billions of connected IoT devices)

Number of global active IoT connections (installed base) in billions



Connectivity type	CAGR	
	20–24	25–35
Cellular 5G/6G IoT	208%	35%
Wireless Neighborhood Area Networks (WNAN)	16%	10%
Wired IoT	4%	4%
Other	20%	12%
LPWA	36%	13%
Cellular IoT (excl. 5G, LPWA)	17%	9%
Wireless Personal Area Networks (WPAN)	7%	9%
Wireless Local Area Networks (WLAN)	16%	10%

XX% = CAGR
(Compound Annual Growth Rate)

Note: IoT connections do not include any computers, laptops, fixed phones, cell phones, or consumer tablets. Counted are active nodes/devices or gateways that concentrate the end-sensors, not every sensor/actuator. Simple one-directional communications technology (e.g., RFID or NFC) is not considered. Wired includes ethernet and field buses (e.g., connected industrial PLCs or I/O modules); The number of wired IoT aggregation nodes represents the primary connection point and excludes all wired end nodes; Cellular includes 2G, 3G, 4G, 5G; LPWAN includes unlicensed and licensed low-power networks; WPAN includes Bluetooth, Zigbee, Z-Wave or similar; WLAN includes Wi-Fi and related protocols; WNAN includes non-short-range mesh, such as Wi-SUN; Other includes satellite and unclassified proprietary networks with any range.

Source: IoT Analytics Research 2025—Global Cellular IoT Connectivity Tracker & Forecast. Conditions for republishing: Source citation with link to original post and company website.

IoT Analytics' prediction

IoT devices **excluding** PC, Laptop, Smartphone, Tablet, and one directional technology like RFID, NFC

Image Source: <https://iot-analytics.com/number-connected-iot-devices/>

Wired: Eth, SCADA network (Modbus, Fieldbus, etc.)

Cellular: 2G, 3G, 4G, 5G; **LPWA:** LoRa, HaLow

WPAN: BLE, ZigBee, Z-Wave, etc.; **WLAN:** WiFi

WNAN: Non-short-range mesh e.g. Wi-SUN

Where is IoT?

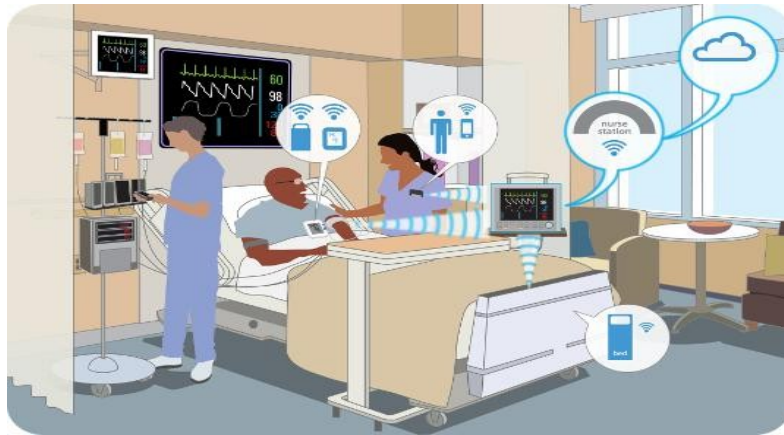


Wearable
Tech Devices



Smart Appliances

It's everywhere!



Healthcare



Industry Automation
and Monitoring

Main Challenges in IoT Implementation

Sensors

- limited resources
- limited types of sensors

Scale

- millions of devices are connected to form IoT

Low Power Network

- devices should remain connected to the network for years
- high network latency
- can't use traditional communication protocols

Interoperability

- various protocol, various architecture
- unavailability of standardized platform
- different technology leads to interoperability issue

Bigdata & Data analytics

- massive amount of sensor data
- different sources and various forms
- extract intelligence from the heaps of data

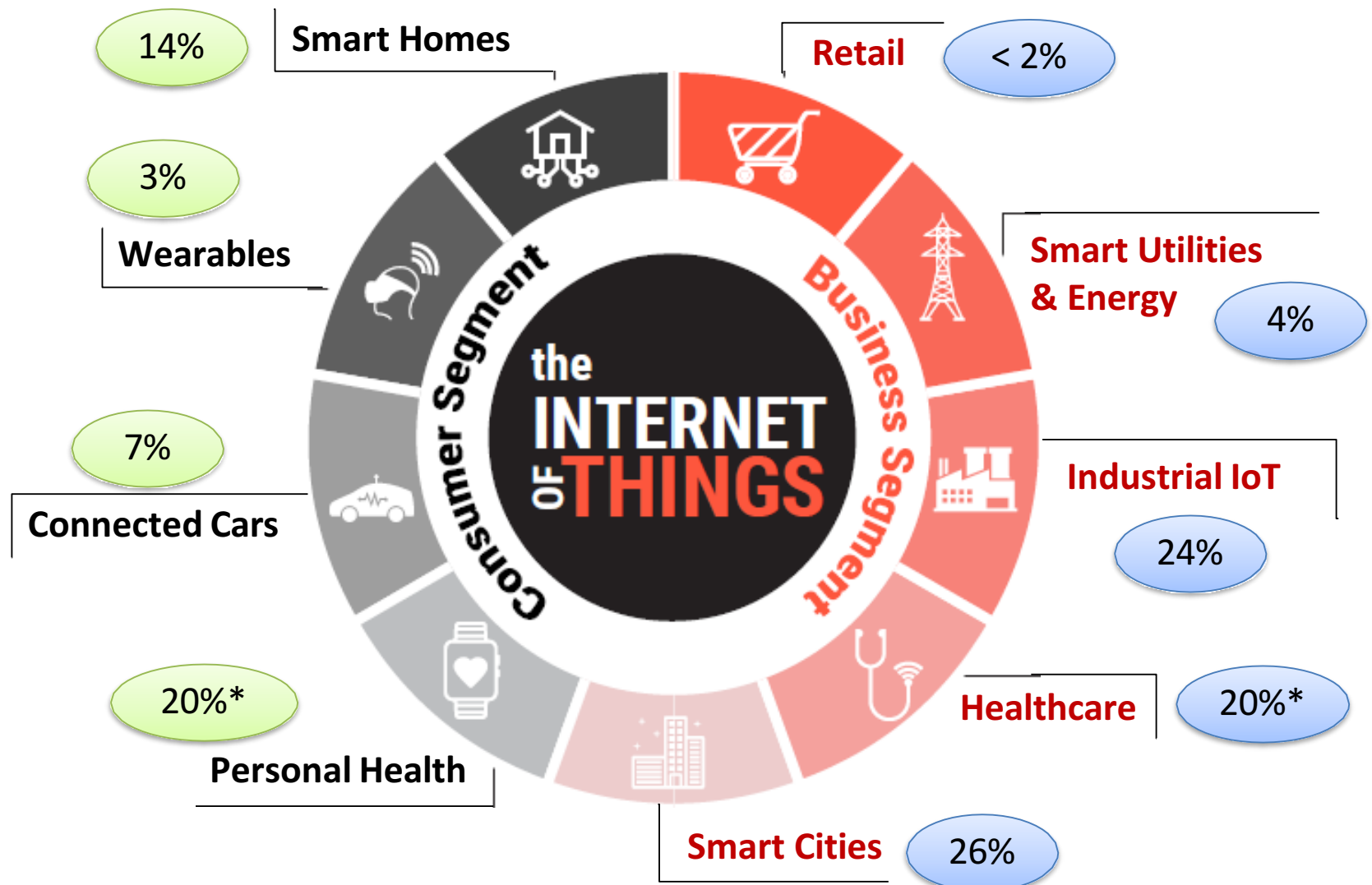
Privacy

- which personal data to share with whom
- how to control

Security

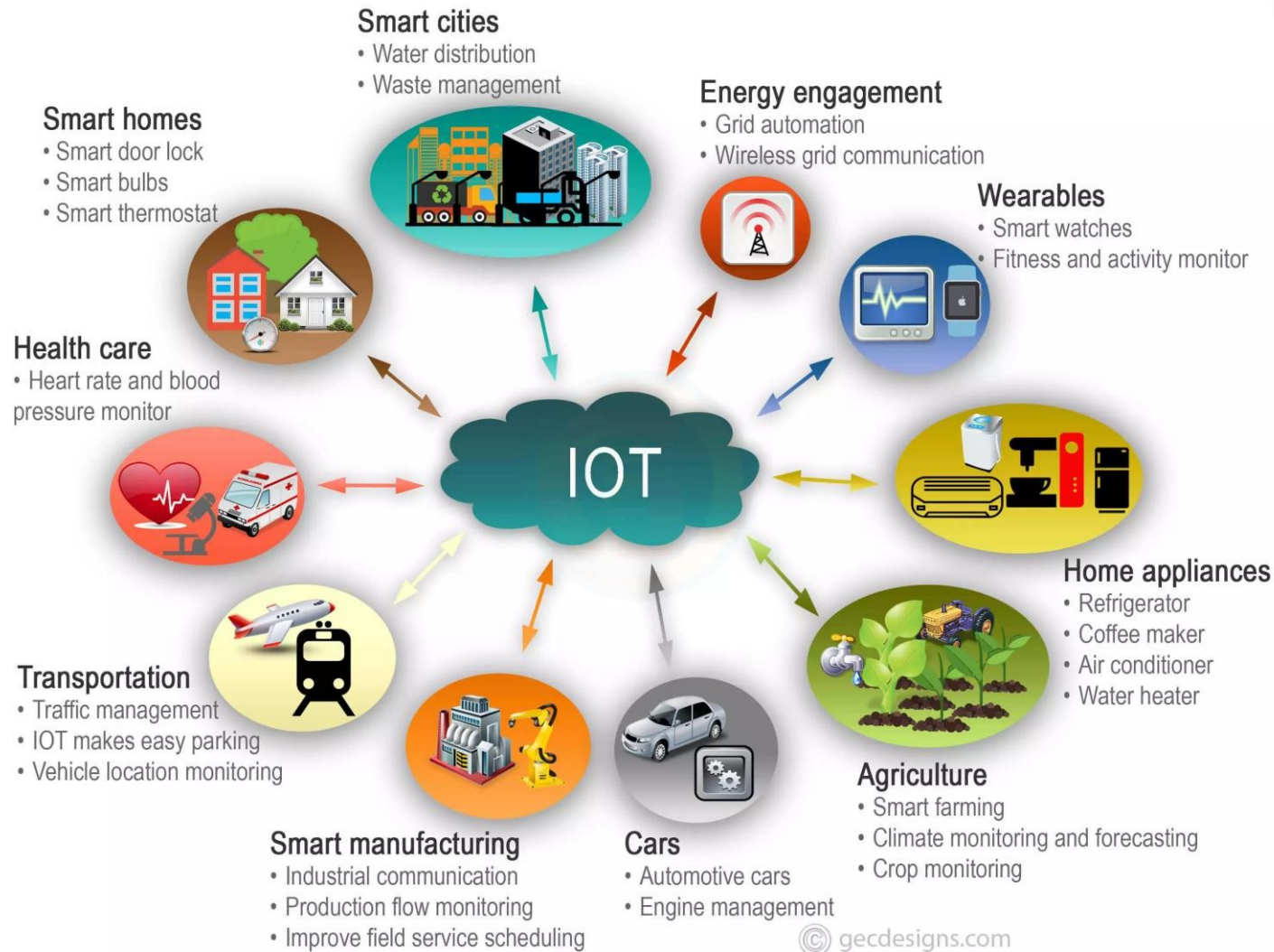
- “things” becomes connected, so security becomes complex

Global IoT Market Share



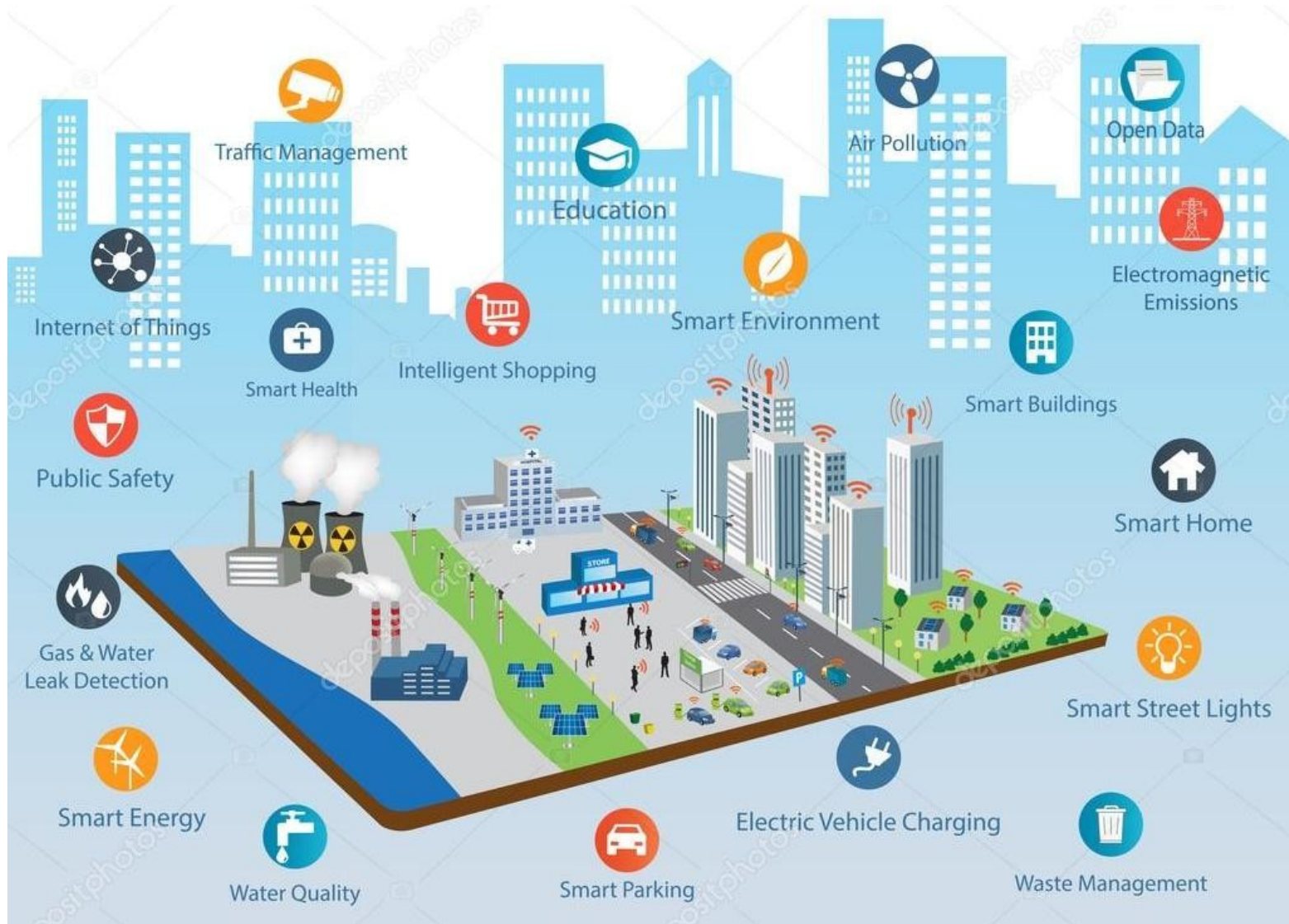
Source: <https://growthenabler.com/flipbook/pdf/IOT%20Report.pdf>

Applications of IoT



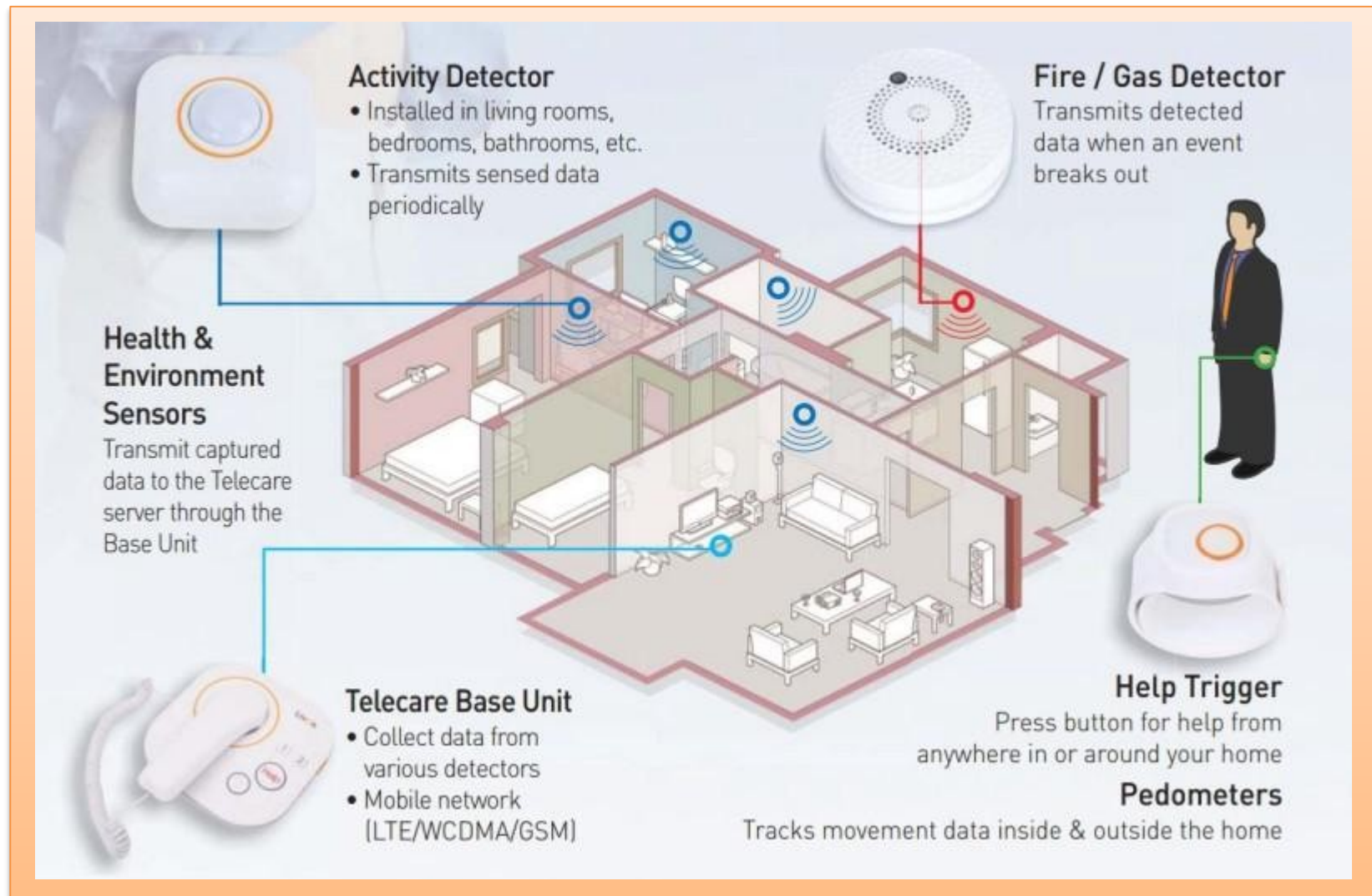
Source: <https://sl.bing.net/jncHHWgAj36>

Smart City



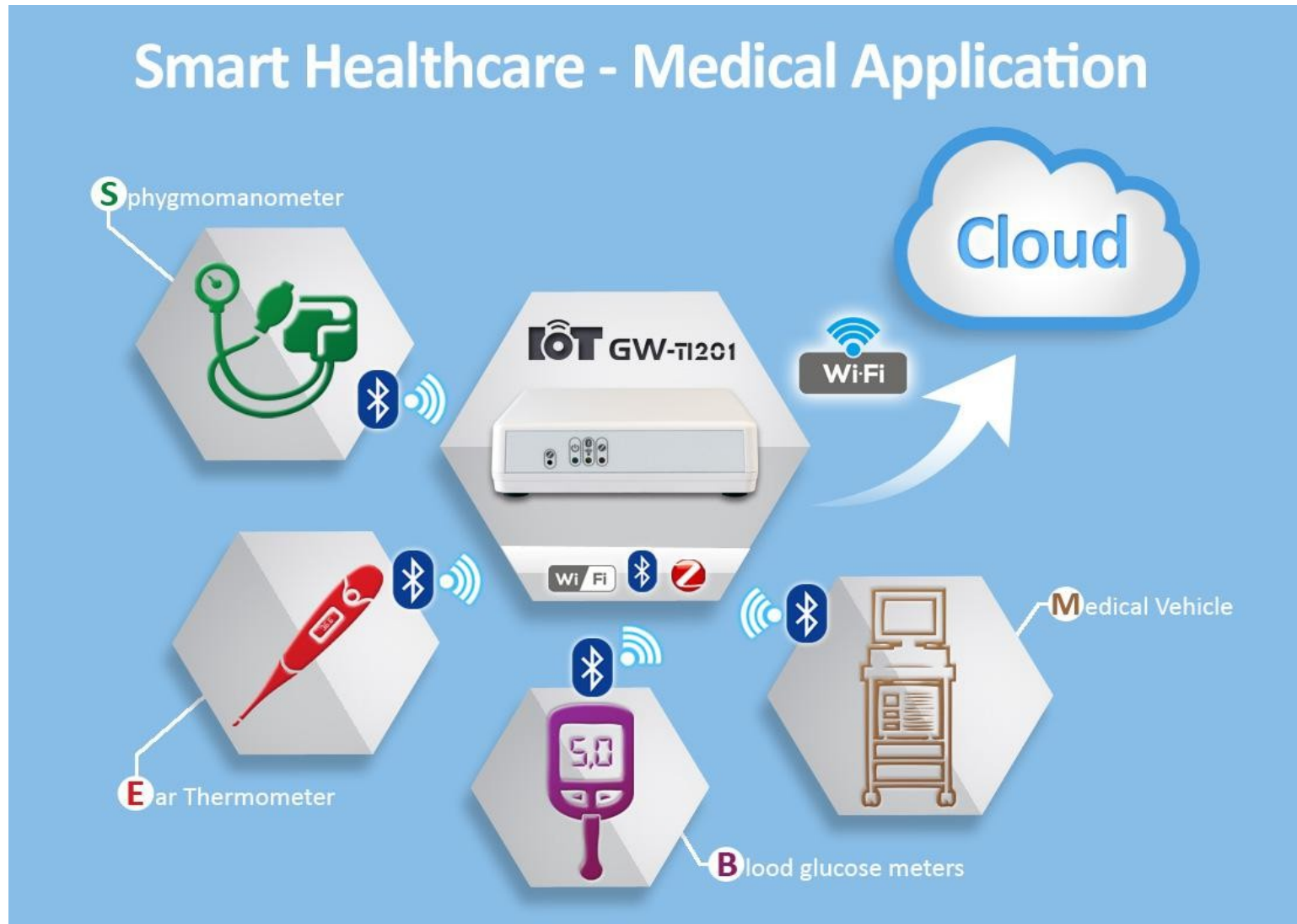
Source: <https://depositphotos.com/126025652/stock-illustration-smart-city-concept-and-internet.html>

Smart Home



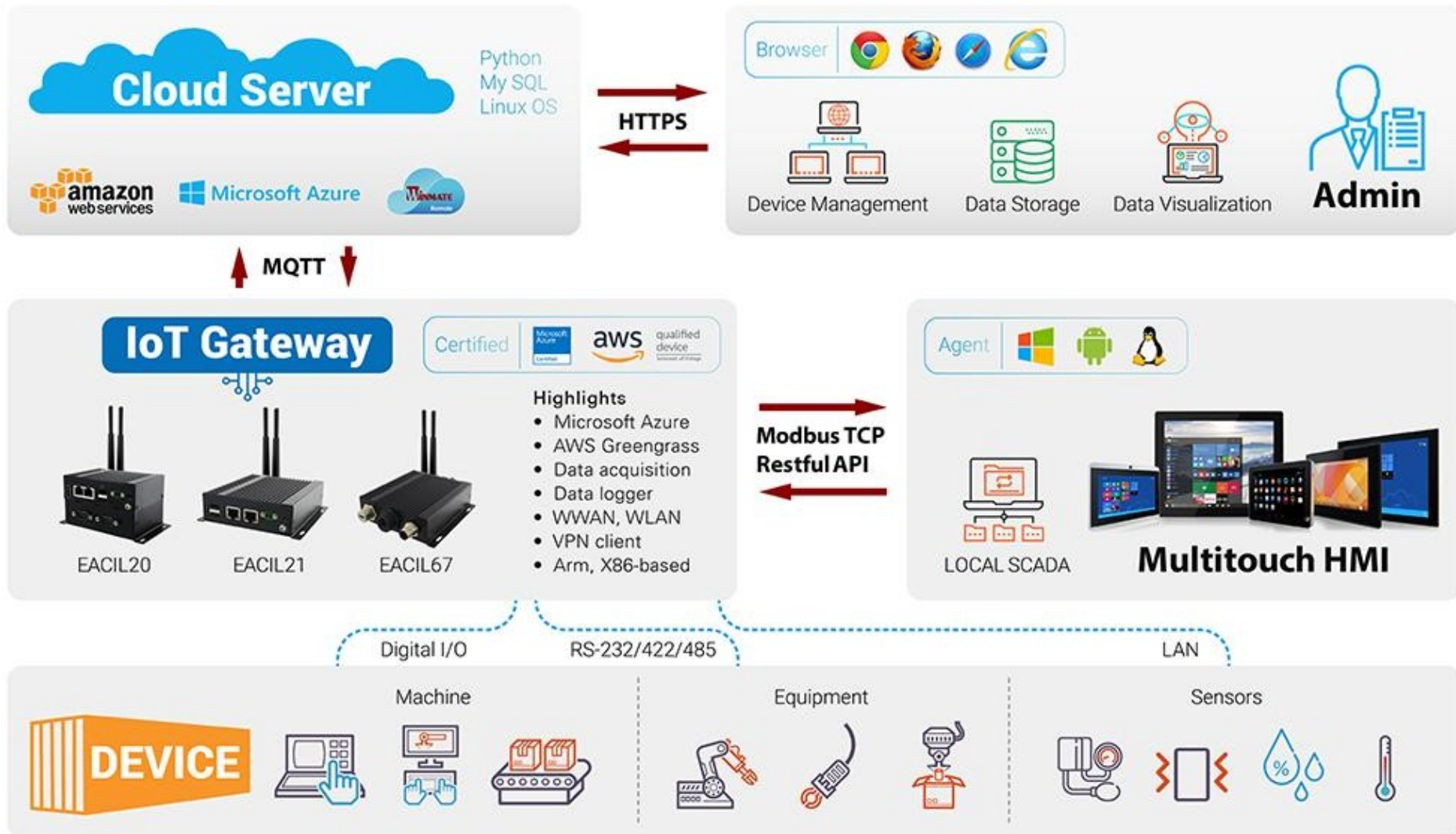
Source: <https://medium.com/@globalindnews/north-america-accounted-for-major-share-in-the-global-smart-home-healthcare-market-in-2015-cc9cc1974ac5>

Smart Healthcare



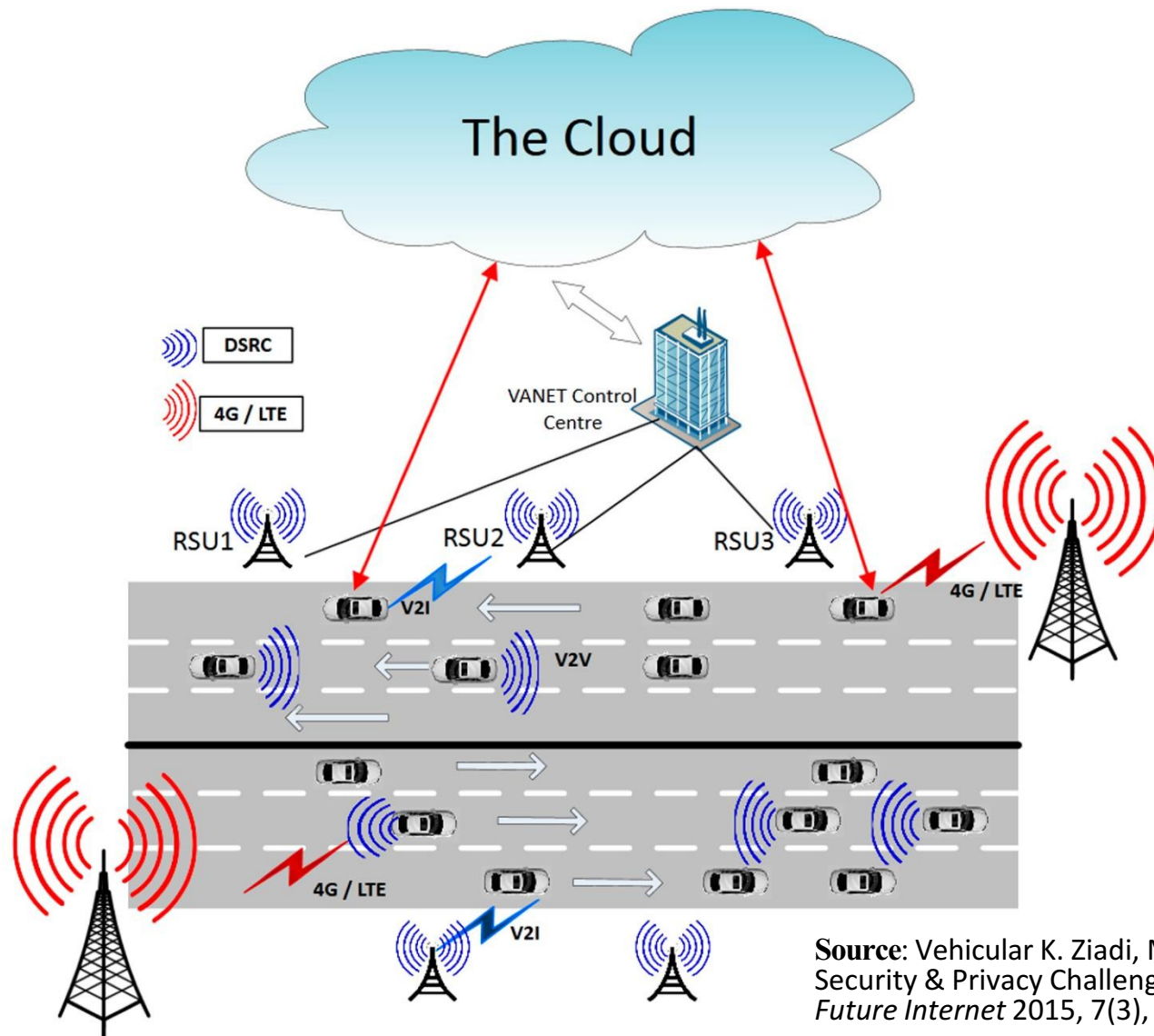
Source: <http://iot.fit-foxconn.com/>

Industrial IoT



Source: https://www.winmate.com/Solutions/Solutions_IoT.asp

Connected Cars



Source: Vehicular K. Ziadi, M. Rajarajan, "Internet: Security & Privacy Challenges and Opportunities", *Future Internet* 2015, 7(3), 257-275.

Google's Self-Driving Car



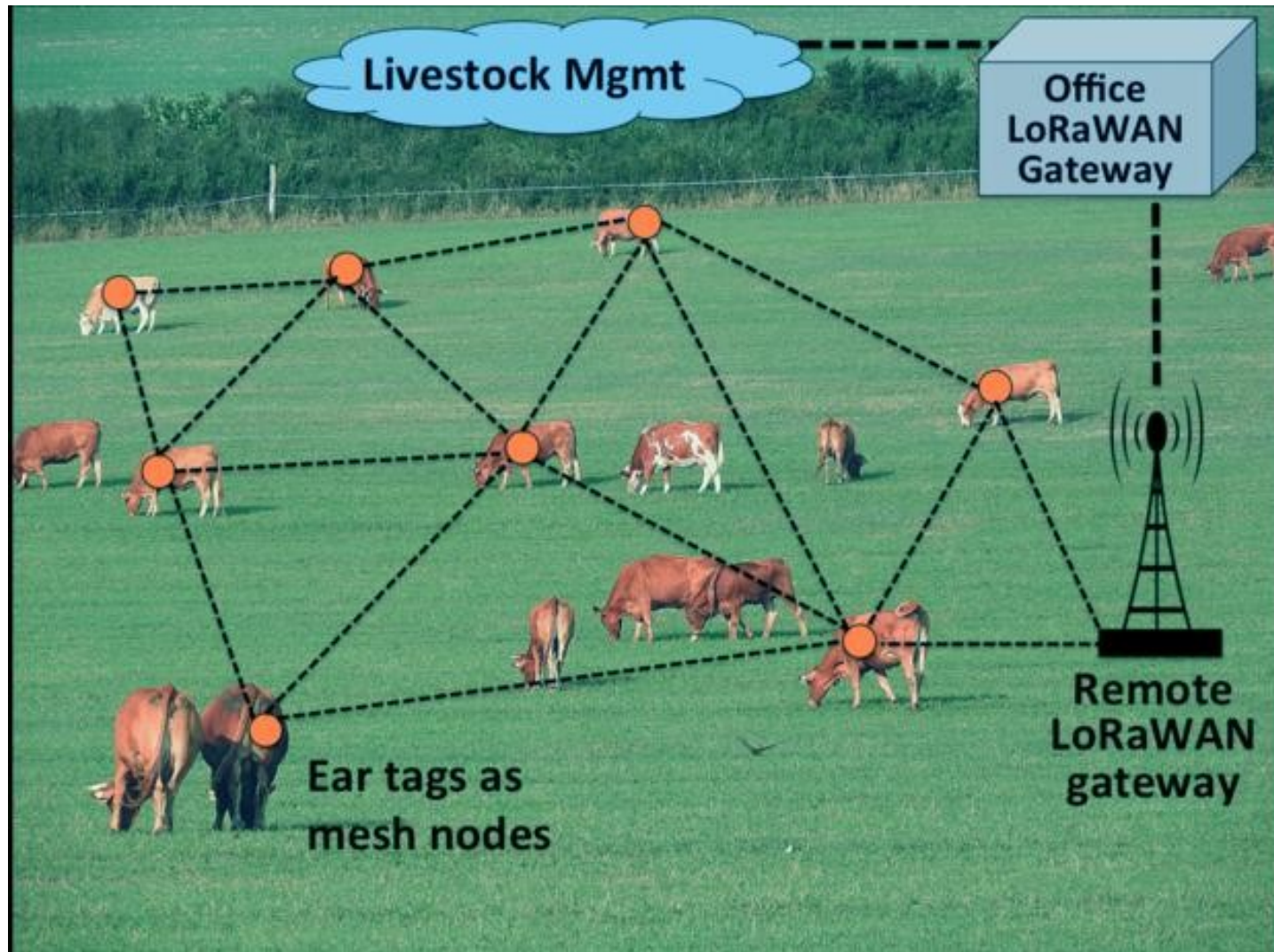
Source: <https://www.google.com/>

Smart Agriculture



Source: <https://in.pinterest.com/pin/515380751093603767/?lp=true>

Livestock Management



Source: <https://data-flair.training/blogs/iot-applications-in-agriculture/>

Traditional Data Flow in IoT

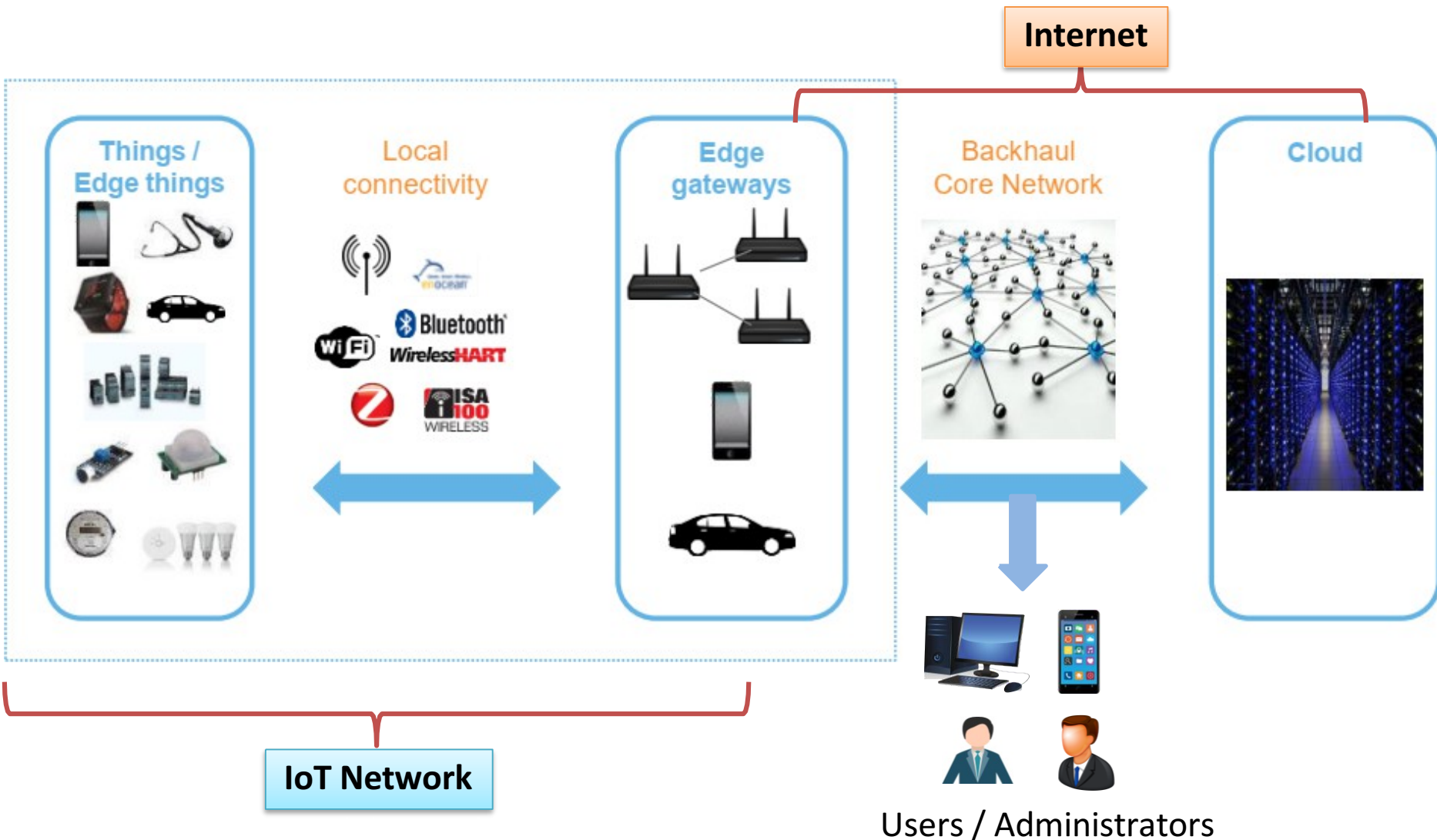
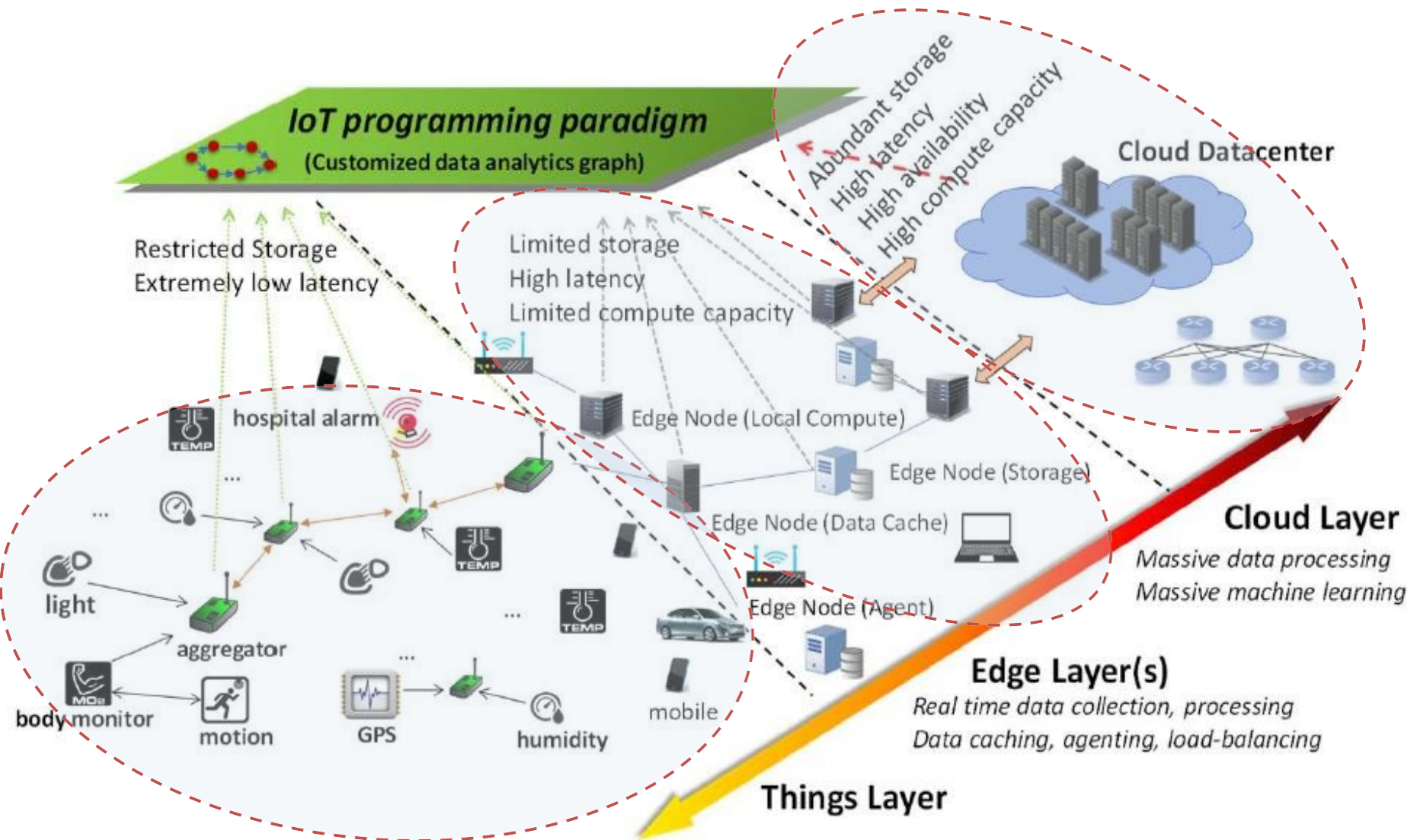


Image Source: <http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf>

Data Processing Locations

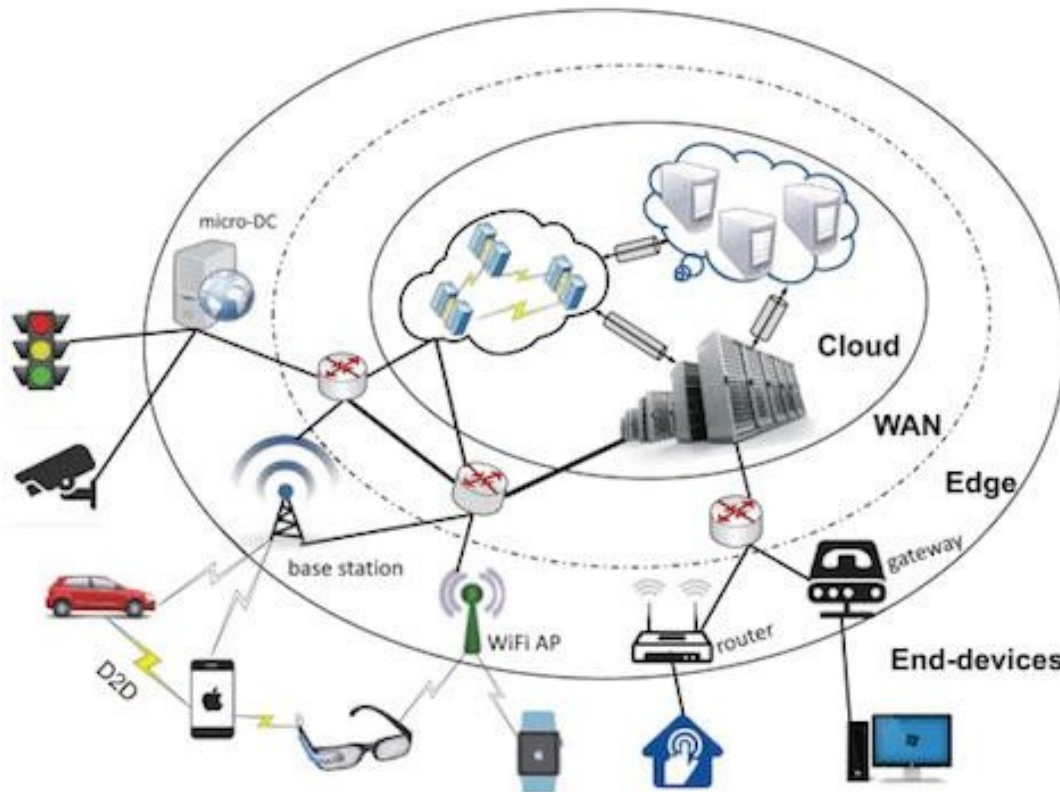


Source: Rajiv Ranjan et. al., "Integrating the IoT and Data Science" IEEE Cloud Computing, 2018

Data Processing at the Edge

Edge Computing

- Host computation tasks close to the data sources and end users.



AI:

- Extracts insights that leads to better decisions & strategic move

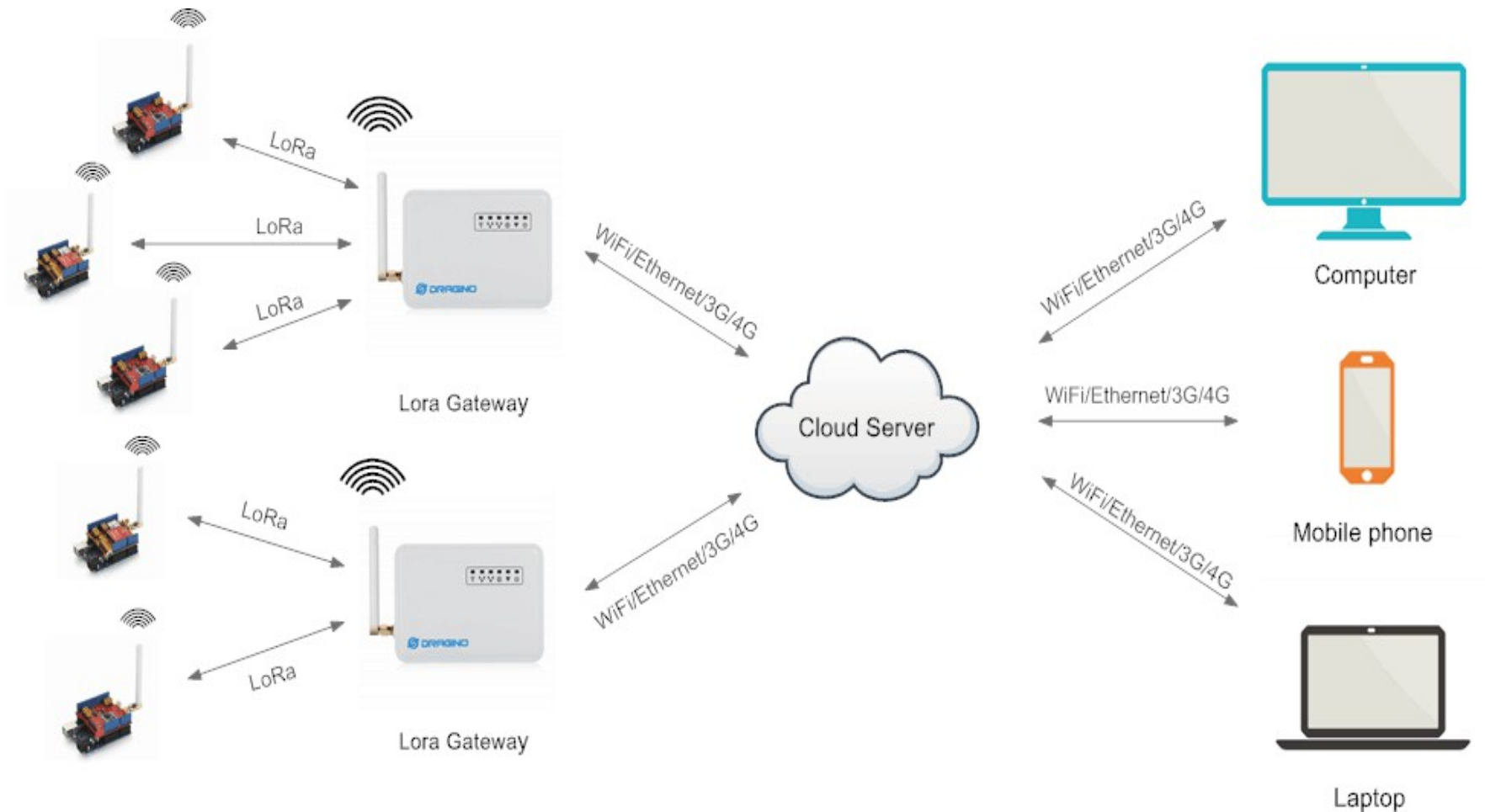
Edge Intelligence:

- Apply AI on data at network Edge

What is needed to know?

- **IoT Components**
 - Application developer needs to know what are the **required components** for designing an IoT application
- **IoT Architecture**
 - For any IoT application it is required to know the **physical arrangement** of IoT sensors, actuators, gateway, edge devices and Internet connectivity
- **IoT Network Access Technology**
 - Types of IoT Network Access Technology or **Connectivity Technology** used to connect the sensors/actuators with each other, and with the IoT gateway or Internet
- **IoT Data Processing / Analysis**
 - Sensed data needs to be **stored and analyzed** to take business decision or to get new business insight. Where it can be done and How?

DIY: Long-range Monitoring & Control



DIY: Smart Home Monitoring

